Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 155612

Subject: Python Programming

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven questions only):*** **[2 x 7 = 14]**

(a) Method overriding is an example of
(i) Compile-time polymorphism (iii) Static dispatch
(ii) Run-time polymorphism. (iv) Abstraction

(b) Which function returns the mean of a dataset in the statistics module?
(i) mean() (iii) avg().
(ii) average() (iv) mid()

(c) Which of the following creates a NumPy array?
(i) array.create() (iii) np.list()
(ii) numpy.make() (iv) np.array() -

(d) What will be output of the program code

```
def test(a, b=[]):
    b.append(a)
    return b
print(test(1))
print(test(2))
```

(i) [1], [2] (iii) [2], [1, 2]
(ii) [1], [1, 2]· (iv) Error

(e) What will be output of the program code

```
class Test:
    __x = 10
t = Test()
print(hasattr(t, '__x'))
```

(i) True (iii) False
(ii) Error (iv) None

(f) What will be printed?

```
d = {'a':1, 'b':2}
print(d.get('c', 5))
```

(i) 5 (iii) KeyError
(ii) 'c' (iv) None

(g) What will the following code print?

```
s = "Python"
print(s[-4:-1])
```

(i) Pyt (iii) yth
(ii) tho· (iv) hon

(h) Which of the following is immutable?
(i) List (iii) Set
(ii) Dictionary (iv) Tuple -

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(i) What is the correct way to merge two dictionaries d1 and d2?
(i) d1 + d2
(ii) d1.append(d2)
(iii) d1.extend(d2)
(iv) d1.update(d2)

(j) Which code finds the sum of all even numbers in a list L?
(i) sum(x for x in L if x % 2 == 0)
(ii) filter(lambda x: x%2==0, L)
(iii) for x in L: if x%2==0: total+=x
(iv) sum(L)

Q.2 (a) Write a Python program that reads integers until a negative number is entered, then displays the sum and count of positive numbers. Explain how the while loop manages the flow and how the termination condition works. [7]

(b) Write a Python program that takes a sentence as input and prints each word on a new line along with its length. Explain why string processing is essential in applications like text analysis. [7]

Q.3 (a) Explain the concepts of function definition, function calling, default parameters, and scope & lifetime of variables in Python. Provide appropriate examples for each concept. [7]

(b) Explain different ways of creating lists in Python. Discuss homogeneous and heterogeneous lists with suitable examples, and describe how lists differ from strings and tuples. [7]

Q.4 (a) Write a Python program to remove all negative numbers from a list using the del statement and list slicing. Show the original and updated lists. [7]

(b) Explain different ways of creating dictionaries in Python. Write a program that creates a dictionary of student names and marks, then adds a new entry, updates an existing entry, and prints the dictionary before and after modification. [7]

Q.5 (a) Explain dictionary methods like keys(), values(), items(), get(), and update() with appropriate examples. Write a program that prints all the key-value pairs separately. [7]

(b) Explain what a frozen set is in Python and how it differs from a normal set. Write a program that converts a list into a frozen set and attempts to modify it (showing the resulting error). [7]

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Q.6 (a) Write a Python program that reads data from one text file, converts all text to uppercase, and writes the result into a new file. Explain how file reading and writing are coordinated in the program. [7]

(b) Explain the concept of serialization in Python using the pickle module. Write a program that pickles (stores) a dictionary into a binary file and then retrieves it back. [7]

Q.7 (a) Explain how the os.path module helps in handling file paths. Write a program using os.path.exists(), os.path.getsize(), and os.path.splitext() to analyze a given file. [7]

(b) Explain exception handling in file operations. Write a program that safely opens a file using try–except–finally and prints an error message if the file does not exist. [7]

Q.8 (a) Define Object-Oriented Programming (OOP). Explain the four major pillars of OOP with appropriate explanations. [7]

(b) Explain the concept of polymorphism in OOP. Discuss both compile-time (overloading) and run-time (overriding) polymorphism with theoretical examples. [7]

Q.9 (a) Explain the difference between Python lists and NumPy arrays. Discuss memory usage, performance, and functionality differences. [7]

(b) Discuss the difference between TensorFlow and Keras. Explain how they work together and why Keras simplifies the model-building process. [7]

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